

$$\begin{cases} -3 \leq x \leq 3 \\ 0 \leq y \leq \sqrt{9-x^2} \\ 0 \leq z \leq \sqrt{9-x^2-y^2} \end{cases}$$

Problem 1. Evaluate the integral using appropriate coordinates:

$$\int_{-3}^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2} dz dy dx.$$

Problem 2. Let W be the region in the first octant lying within the sphere $x^2 + y^2 + z^2 = 1$ and between the two vertical planes $y = x$ and $y = \frac{1}{\sqrt{3}}x$. Evaluate the integral using appropriate coordinates:

$$\iiint_W x dV.$$

$$\begin{aligned} x &= \rho \sin \phi \cos \theta \\ y &= \rho \sin \phi \sin \theta \\ z &= \rho \cos \phi \end{aligned}$$

Problem 3. Let D be the parallelogram enclosed by the lines $x - 2y = 0$, $x - 2y = 4$, $3x - y = 1$, and $3x - y = 8$. We shall calculate the following integral using the change of variables formula:

$$\iint_D \frac{x-2y}{3x-y} dA.$$

$$\text{Jac} = \rho^2 \sin \phi$$

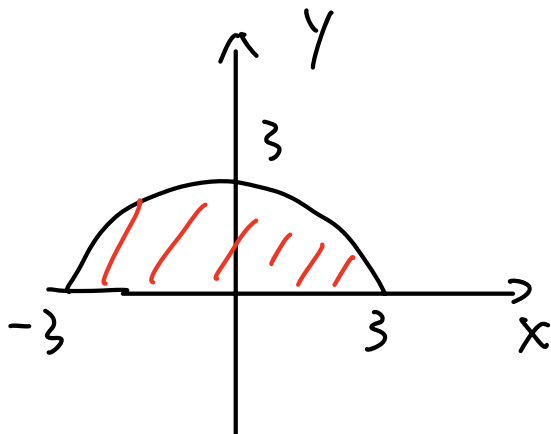
- Sketch the region D .
- Make a guess at what u and v should be as functions in terms of x, y (so $u = u(x, y)$ and $v = v(x, y)$). If your group is unsure about your choice, have the TA check your answer before proceeding.
- Sketch the region D_0 in the uv -plane that corresponds to the region D in the xy -plane.
- Use the change of variables formula to calculate the double integral in the problem statement.

Problem 1

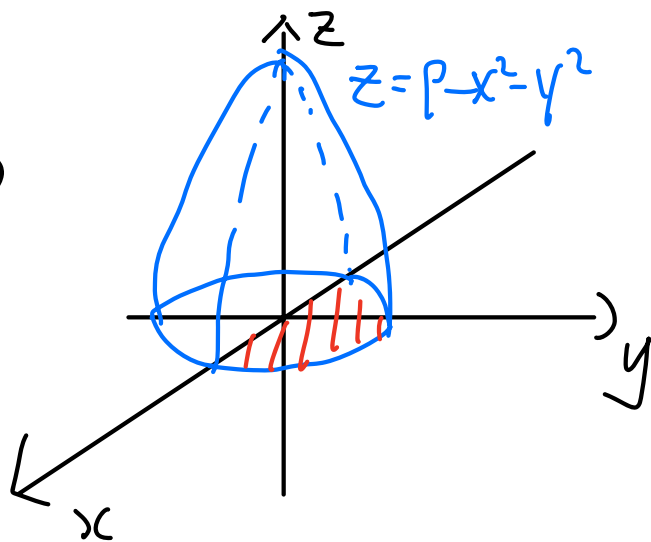
$$-3 \leq x \leq 3$$

$$0 \leq y \leq \sqrt{9-x^2} \Rightarrow$$

$$0 \leq z \leq 9-x^2-y^2$$



$\Rightarrow$



Using cylindrical coordinates, the integral is then

$$\int_0^\pi \int_0^3 \int_0^{9-r^2} \underbrace{\sqrt{x^2+y^2}}_{r \cdot r} \underbrace{dz}_{\text{Jacobian factor}} dr d\theta$$

$$= \int_0^\pi \int_0^3 r^2(9-r^2) dr d\theta$$

$$= \int_0^\pi \left(3r^3 - \frac{1}{5}r^5 \right) \Big|_0^3 d\theta$$

$$= \pi \cdot \left(81 - \frac{3^5}{5} \right) = \frac{162}{5} \pi$$



Problem 2

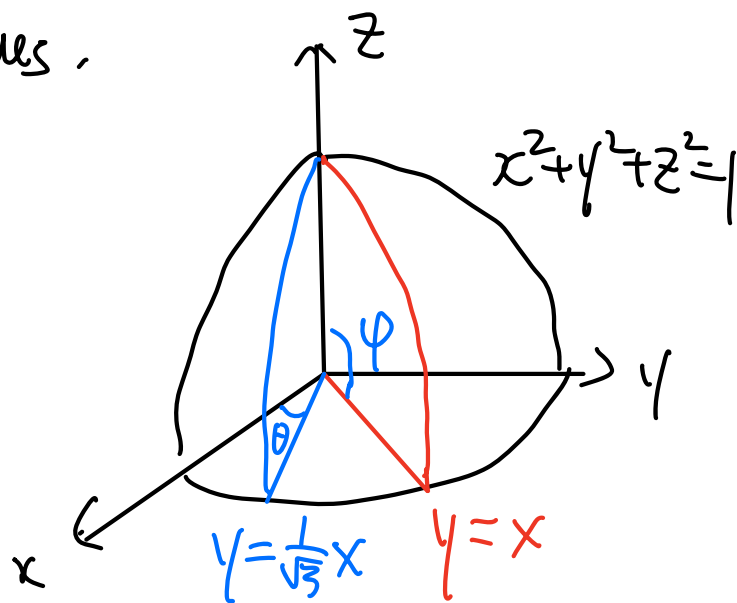


Using spherical coordinates,

$$x = \rho \sin \varphi \cos \theta$$

$$y = \rho \sin \varphi \sin \theta$$

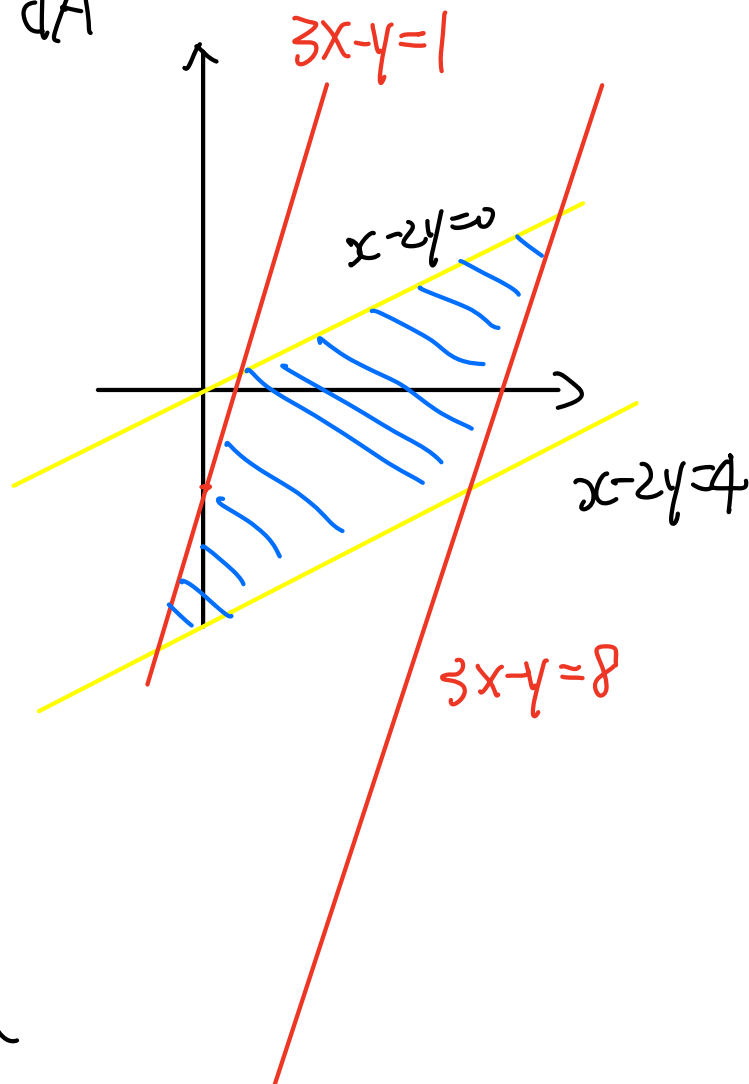
$$z = \rho \cos \varphi$$



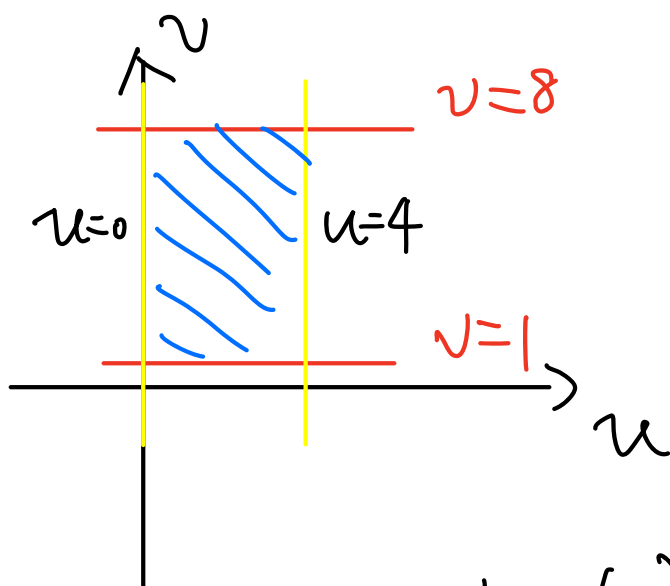
$$\begin{aligned}
 & \iiint_W x \, dV \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \int_0^{\frac{\pi}{2}} \int_0^1 \underbrace{\rho \cdot \sin \varphi \cos \theta}_x \cdot \underbrace{\rho^2 \cdot \sin \varphi}_{\text{Jac}} \, d\rho \, d\varphi \, d\theta \\
 &= \left(\int_0^1 \rho^3 \, d\rho \right) \cdot \left(\int_0^{\frac{\pi}{2}} \sin^2 \varphi \, d\varphi \right) \left(\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos \theta \, d\theta \right) \\
 &= \frac{1}{4} \cdot \frac{1}{4} \pi \cdot \frac{\sqrt{2}-1}{2} = \frac{\sqrt{2}-1}{32} \pi
 \end{aligned}$$

Problem 3 $\iint_D \frac{x-2y}{3x-y} dA$

(b) $u = x - 2y$ $v = 3x - y$
 (or equivalently.
 $u = 3x - y$, $v = x - 2y$)



(c)



(d) $J(u, v) = \det \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$
 $= \det \begin{pmatrix} -\frac{1}{5} & -\frac{3}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix} = \frac{1}{5}$

$u = x - 2y \Rightarrow x = \frac{1}{5}(2v - u)$
 $v = 3x - y \Rightarrow y = \frac{1}{5}(v - 3u)$

$\iint_D \frac{x-2y}{3x-y} dA = \int_1^8 \int_0^4 \frac{u}{v} \cdot \left| \frac{1}{5} \right| du dv = \frac{8}{5} \ln(8)$